Dictionary for `phonon_coupling_report_{i}.txt`

Scores used in the report

Singular values σ_i (report fields: $\sigma_{\min}$, σ_{mean}) Given two subspaces with orthonormal bases Q_A, Q_B,

$$C = Q_A^\dagger Q_B, \qquad \{\sigma_i\} = \text{svd}(C), \qquad 0 \le \sigma_i \le 1.$$

Interpretation: $\sigma_i \approx 1$ means aligned; $\sigma_i \approx 0$ means orthogonal.

Principal angles θ_i (report fields: θ_{mean}, $\theta_{\max}$)

$$\theta_i = \arccos(\sigma_i) \times \frac{180}{\pi} \quad \text{(degrees)}.$$

Interpretation: 0° is aligned; 90° is orthogonal.

k_{DFT} **(`k_dft`)** $k_{\text{DFT}} = \dim\left(\text{span}\{\mathbf{e}_m^{\text{DFT}} : m \in S_\tau^{\text{DFT}}\}\right)$. Interpretation: dimension of the DFT coupling-mode subspace.

k_{ML} **(`k_ml`)** analogous dimension for the MLIP coupling-mode subspace. Interpretation: if $k_{\text{ML}} < k_{\text{DFT}}$, the subspace score X is capped.

X **(`AvgProjPowX`, printed as `X`)** For coupling-mode subspaces Q_{DFT} and Q_{ML},

$$X(\mathbf{q}) = \frac{1}{k_{\text{DFT}}} \sum_{i=1}^{\min(k_{\text{DFT}}, k_{\text{ML}})} \sigma_i^2 \in [0, 1].$$

Interpretation: coupling-subspace agreement; closer to 1 is better.

`X: mean/min/max` For multiple q-points: `mean` $= \langle X(\mathbf{q})\rangle_{\mathbf{q}}$ and `min/max` are extrema across analyzed $\mathbf{q}$. Interpretation: variability of subspace agreement over q (with Γ-only, these often coincide).

X_{mean} **(`X_mean`)** $X_{\text{mean}} = \langle X(\mathbf{q})\rangle_{\mathbf{q}}$.

`dQ score: mean/min` In the per-model header, (see below for `Score()`)

$$\texttt{mean} = \langle \text{Score}(\mathbf{q})\rangle_{\mathbf{q}} = \text{Score_mean}, \qquad \texttt{min} = \min_{\mathbf{q}} \text{Score}(\mathbf{q}).$$

`L1w` and `L1w_mean`

$$\text{L1w}(\mathbf{q}) = \sum_{C \in \mathcal{R}_\tau(\mathbf{q})} \left| w_C^{\text{ML}}(\mathbf{q}) - w_C^{\text{DFT}}(\mathbf{q}) \right|.$$

Interpretation: cluster-weight mismatch in DFT-important spectral regions; 0 is perfect.

`L1w_mean` L1w_mean $= \langle \text{L1w}(\mathbf{q})\rangle_{\mathbf{q}}$. Interpretation: average L1 cluster-weight mismatch across analyzed q-points.

`wtheta_mean` wθ_mean $= \langle w\theta(\mathbf{q})\rangle_{\mathbf{q}}$. Interpretation: average DFT-weighted cluster misalignment angle (degrees).

`w(1-sigma2)_mean` $\langle w(1-\sigma^2)(\mathbf{q})\rangle_{\mathbf{q}}$. Interpretation: average cluster-level σ^2 misalignment score across q-points.

`wtheta` **and** `wtheta_mean` For each relevant cluster C, compute principal angles between Q_C^{DFT} and $Q_{W(C)}^{\mathrm{ML}}$ and take the mean angle $\overline{\theta}_C(\mathbf{q})$. Then

$$w\theta(\mathbf{q}) = \sum_{C \in \mathcal{R}_\tau(\mathbf{q})} w_C^{\mathrm{DFT}}(\mathbf{q})\, \overline{\theta}_C(\mathbf{q}) \quad \text{(degrees)}.$$

Interpretation: DFT-weighted subspace misalignment inside the important frequency windows; smaller is better.

`w(1-sigma2)` **and** `w(1-sigma2)_mean` For each relevant cluster C, define

$$\sigma_C^2(\mathbf{q}) = \frac{1}{\dim(Q_C^{\mathrm{DFT}})} \sum_i \sigma_i(C;\mathbf{q})^2, \qquad w(1-\sigma^2)(\mathbf{q}) = \sum_{C \in \mathcal{R}_\tau(\mathbf{q})} w_C^{\mathrm{DFT}}(\mathbf{q})\big(1-\sigma_C^2(\mathbf{q})\big).$$

Interpretation: another cluster-level misalignment score; 0 is best.

O_{ij} **(used inside** `E_vec`**)**

$$O_{ij}(\mathbf{q}) = \left|\mathbf{e}_i^{\mathrm{DFT}}(\mathbf{q})^\dagger \mathbf{e}_j^{\mathrm{ML}}(\mathbf{q})\right|^2 \in [0,1].$$

Interpretation: phase/sign-invariant similarity between individual modes.

π **(Hungarian assignment, used inside** `E_freq`**,** `E_vec`**)**

$$\pi = \arg \max_{\text{one-to-one maps}} \sum_{i=1}^{3N} O_{i,\pi(i)}.$$

Interpretation: matches MLIP modes to DFT modes without trusting mode indices (or frequencies)

`weight_kind` Selects nonnegative DFT importance weights w_i for the targeted (mode-by-mode) RMS scores.

$$w_i \propto |a_i|^2$$

Interpretation: emphasizes modes that contribute to the GS→ES displacement, or, depending on kind, emphasises higher frequency modes.

α (`alpha`) Tradeoff scalar in the final targeted score. Interpretation: larger α penalizes eigenvector mismatch more strongly relative to frequency error. I settled for around 1.3.

`E_freq` With normalized weights $\tilde{w}_i = w_i / \sum_j w_j$ and assigned partner $\pi(i)$,

$$E_{\mathrm{freq}}(\mathbf{q}) = \sqrt{\sum_i \tilde{w}_i \left(\omega_{\pi(i)}^{\mathrm{ML}}(\mathbf{q}) - \omega_i^{\mathrm{DFT}}(\mathbf{q})\right)^2}.$$

Interpretation: weighted RMS frequency error on GS→ES- DFT coupling modes; 0 is best.

`E_freq_rel`

$$E_{\mathrm{freq,rel}}(\mathbf{q}) = \sqrt{\sum_i \tilde{w}_i \left(\frac{\omega_{\pi(i)}^{\mathrm{ML}} - \omega_i^{\mathrm{DFT}}}{\max(|\omega_i^{\mathrm{DFT}}|, \varepsilon)}\right)^2}.$$

Interpretation: weighted RMS *relative* frequency error (dimensionless).

`E_vec` Define $d_i(\mathbf{q}) = 1 - O_{i,\pi(i)}(\mathbf{q}) \in [0,1]$. Then

$$E_{\mathrm{vec}}(\mathbf{q}) = \sqrt{\sum_i \tilde{w}_i\, d_i(\mathbf{q})^2}.$$

Interpretation: weighted RMS displacement-pattern mismatch; 0 is best.

`Score` **and** `Score_mean`

$$\mathrm{Score}(\mathbf{q}) = E_{\mathrm{freq}}(\mathbf{q}) + \alpha\, E_{\mathrm{vec}}(\mathbf{q}), \qquad \mathrm{Score_mean} = \langle \mathrm{Score}(\mathbf{q}) \rangle_{\mathbf{q}}.$$

Interpretation: used for ranking MLIPs (smaller is better).

`q-points used` n_q = number of q-points included in the report. Interpretation: with `Gamma_only`=`True`, $n_q = 1$, so "mean/min/max" coincide.